

Numerical Resolution of the Five-Dimensional Three-Mode NESS Density

NLS numerical study

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Abstract

We construct an absolutely normalized numerical steady-state density for the driven three-mode chain in the complete reduced state $x = (I_1, I_2, I_3, \theta_1, \theta_3)$. The final estimator passes independent trajectory-level validation and blind tests at $\Delta t = 2.5 \times 10^{-4}$, the exact equilibrium known-answer control, and an independently fitted replication at $\Delta t = 10^{-3}$. Fine/coarse centered log-density RMS is 0.0243 on both test supports, below the frozen 0.10 gate. The warranted claim is a numerical solution on sampled stationary support, not an analytic global solution or a guarantee in unsampled tails.

1 Question and claim boundary

The goal is stronger than plotting stationary marginals: it is to provide a callable approximation to the full five-dimensional density $\rho_{\text{ss}}(I_1, I_2, I_3, \theta_1, \theta_3)$, including an absolute normalization. We call the task successful only if independent held-out trajectories satisfy all preregistered density, stationarity, robustness, and timestep gates. “Resolved” below therefore means resolved on the region sampled by the stationary trajectories. It does not mean that the learned tail outside that region has been verified.

2 Dynamics, data, and density representation

The driven case uses $(T_1, T_3) = (2, 8)$ and $\gamma = 0.1$; the known-answer control uses $(5, 5)$. The controlled Cartesian integrator is projected to the reduced state only when snapshots are recorded. At the final fine step, $\Delta t = 2.5 \times 10^{-4}$, 64 independent streams per case are generated with burn-in 500, stationary duration 1000, and snapshot interval 0.01. Every stream has 100001 rows. The 32 validation and 32 test streams are assigned by a frozen hash before any physical statistic is read. All actions are positive, all entries finite, and projection, floor, zero-radius, midpoint-failure, and non-finite counters are zero. A separate 64-stream data set supplies the coarse replication at $\Delta t = 10^{-3}$.

In the reduced energy convention used by the solver,

$$E(x) = \frac{1}{2}M^2 - \frac{1}{4} \sum_{j=1}^3 I_j^2 + I_1 I_2 \cos \theta_1 + I_2 I_3 \cos \theta_3, \quad M = I_1 + I_2 + I_3. \quad (1)$$

We anchor the density to the exact $T = 5$ equilibrium form,

$$\rho_m(x) = \frac{\exp[-E(x)/5 + f_m(x)]}{Z_{\text{eq}} Z_m}, \quad \hat{\rho}_{\text{ss}}(x) = \frac{1}{3} \sum_{m=1}^3 \rho_m(x). \quad (2)$$

The three corrections f_m are independently seeded two-layer, 128-unit tanh networks with log-action, periodic-angle, and explicit transport features. The Gibbs anchor enforces physically appropriate action-tail decay. Training combines the stationary Fokker–Planck residual and twelve moment constraints. Complete trajectories, never individual snapshots, define the training, validation, test, and bootstrap units.

The equilibrium partition is evaluated independently as

$$Z_{\text{eq}} = 514.3433631455, \quad \text{SE}(Z_{\text{eq}}) = 1.1432 \times 10^{-5}. \quad (3)$$

The ratio normalizers Z_m are estimated on held-out equilibrium-reference streams. Their values are

$$\begin{aligned} (\log Z_1, \log Z_2, \log Z_3)_{\text{fine}} &= (-0.0081525, -0.0040901, 0.0019366), \\ (\log Z_1, \log Z_2, \log Z_3)_{\text{coarse}} &= (-0.0015192, 0.0021328, 0.0053924). \end{aligned}$$

Additive offsets of individual neural corrections are therefore removed before ensemble averaging.

3 Frozen validation gates

Every model must have finite un-clipped weights, ESS fraction at least 0.10, median and 90th-percentile absolute relative stationary-equation residual at most 0.10 and 0.50, all twelve stream-bootstrap moment discrepancies below $|z| = 3.7482866$, and all one- and two-angle marginal total-variation distances below 0.05. Across the three seeds, the largest centered log-ratio RMS must be below 0.10. The exact equilibrium ratio $f = 0$ is subjected to the same pipeline. A test split is opened only after its validation stage passes.

4 Independent fine-timestep result

Table 1: Driven fine-step blind-test diagnostics.

seed	ESS	FP median	FP p90	max $ z $	max TV
43201	0.9160	0.08875	0.4060	1.343	0.02429
43202	0.9165	0.09501	0.4087	1.443	0.02440
43203	0.9178	0.09141	0.3965	1.220	0.02425

All fine-step gates pass. The maximum pairwise centered log-ratio RMS is 0.01273. The exact equilibrium test also passes: FP residuals are at floating point zero, maximum $|z|$ is 1.272, and maximum TV is 0.03321.

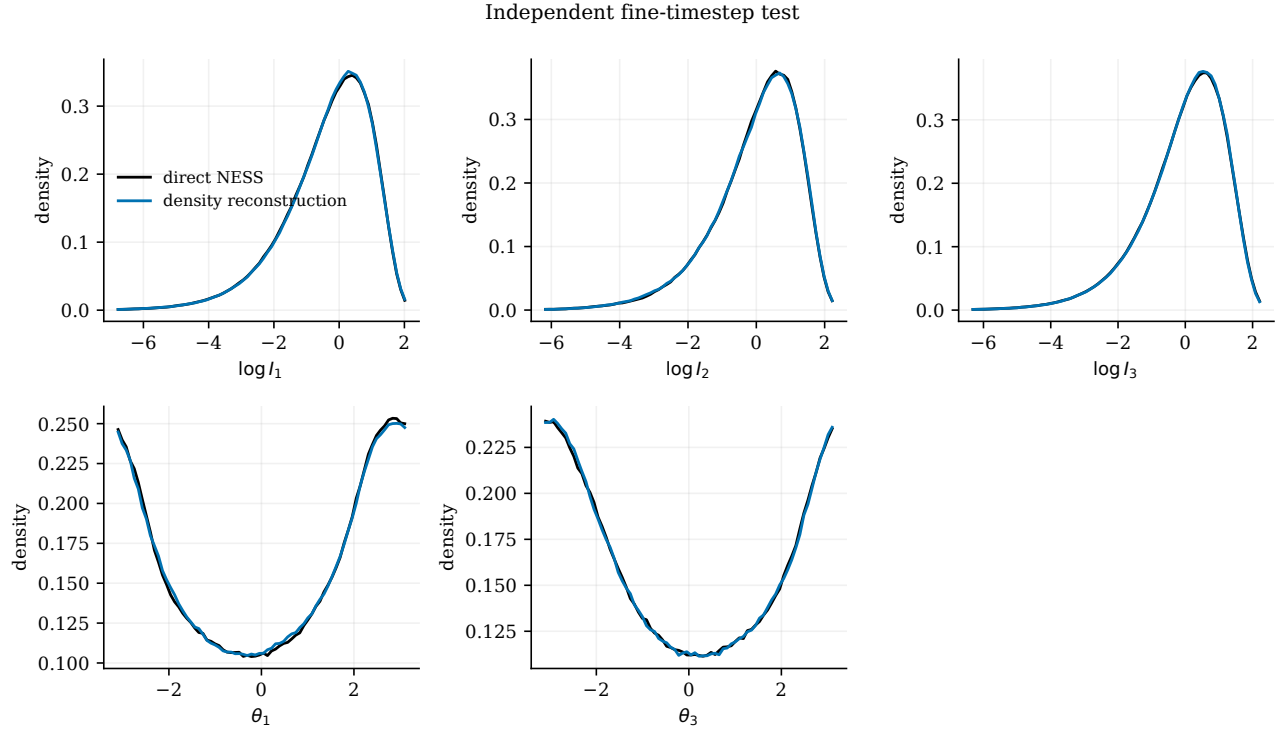


Figure 1: Independent fine-step test marginals. Black: direct NESS. Blue: normalized density reconstruction; the band spans three training seeds.

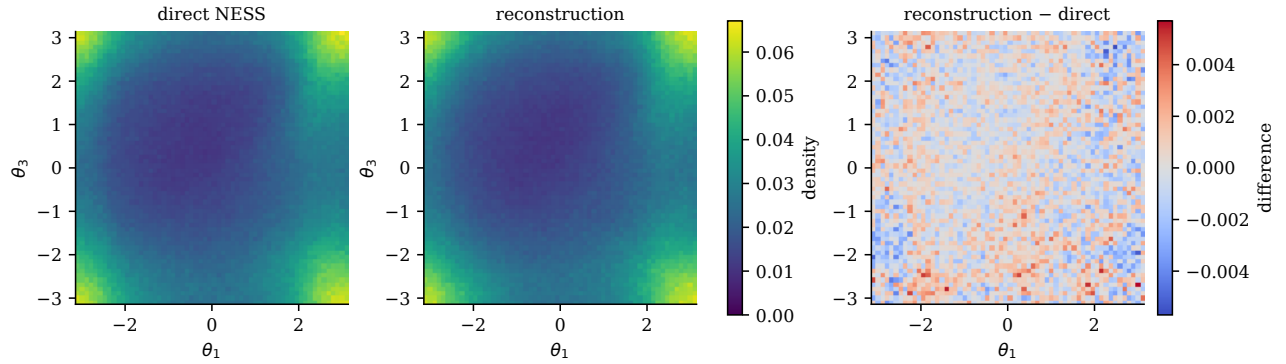


Figure 2: Independent fine-step joint angular marginal and residual.

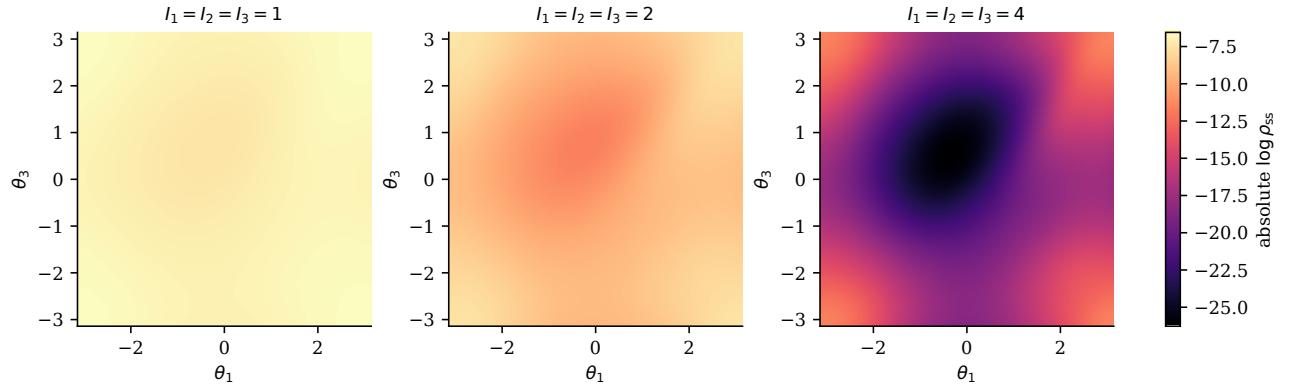


Figure 3: Absolute log-density slices at three equal-action levels.

5 Coarse replication and transparent recovery

The initially frozen two-moment coarse replication did not pass validation. For seed 45202 the right-bond transport moment $I_2 I_3 \sin \theta_3$ had $|z| = 3.9628$, exceeding 3.7483. No coarse test was opened. This failed result is retained under `formal_v7_coarse`.

In a separately frozen recovery, the right-bond moment was added as a third exponential calibration observable beside I_1 and $I_1 I_2 \sin \theta_1$. The neural weights were unchanged and the inherited 0.625 shrink factor was fixed in advance. Validation then passed, authorizing one test access.

Table 2: Driven coarse-step one-shot test diagnostics after R10 recovery.

seed	ESS	FP median	FP p90	max $ z $	max TV
46201	0.9087	0.09121	0.4089	2.830	0.02409
46202	0.9096	0.08943	0.4022	2.695	0.02405
46203	0.9114	0.09310	0.4345	2.652	0.02419

All coarse gates pass, with pairwise RMS 0.01383. The coarse equilibrium known-answer test also passes, with maximum $|z| = 1.417$, maximum TV 0.03360, and machine-zero stationary residual.

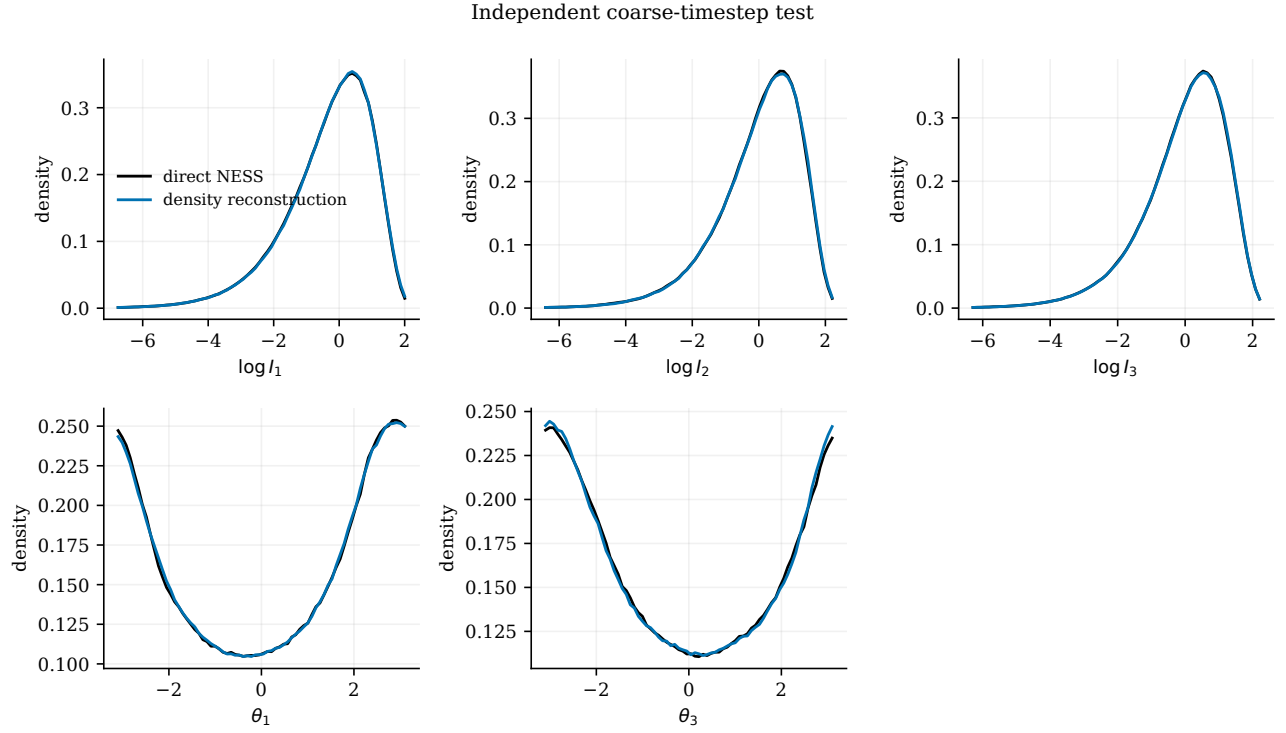


Figure 4: Independent coarse-step test marginals.

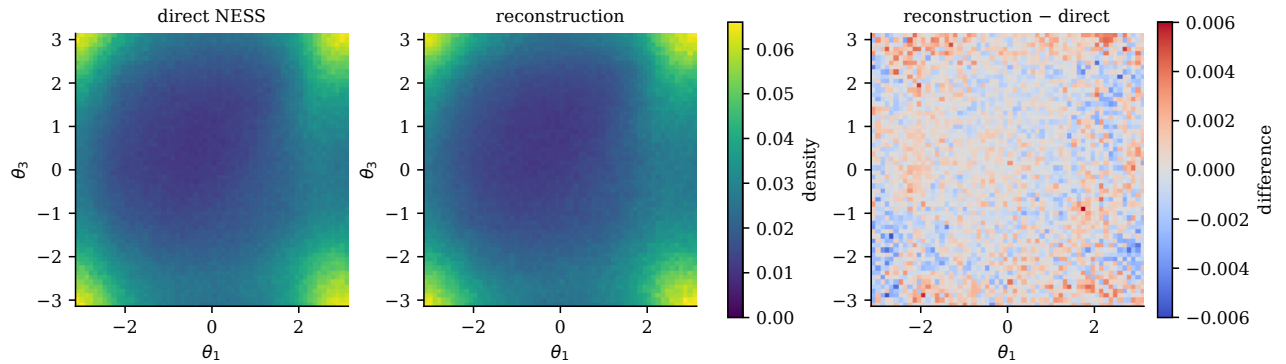


Figure 5: Independent coarse-step joint angular marginal and residual.

6 Timestep consistency

The fine and coarse ensembles are compared after removing the physically irrelevant additive log-density constant. On 50000 deterministic test states from each support, the ensemble centered log-density RMS values are

$$0.024257 \quad (\text{fine support}), \quad 0.024272 \quad (\text{coarse support}). \quad (4)$$

Both are well inside the frozen 0.10 bound. All three paired-seed comparisons also pass, ranging from 0.0258 to 0.0269.

7 Verdict

The complete reduced five-dimensional NESS density is numerically resolved on sampled stationary support. The conclusion rests jointly on fresh held-out trajectory tests, a stationary-equation residual, twelve observables, one- and two-angle marginals, three independent seeds, exact equilibrium controls, and timestep replication. The callable implementation returns a finite positive absolute density and is distributed with the model bundles.

The claim is deliberately narrower than an exact solution: the network defines a smooth extension over positive actions and periodic angles, but accuracy in extreme regions not represented by stationary samples is untested. Therefore the result should not be described as an analytic proof or as pointwise global control of $L^\dagger \rho = 0$.

Reproducibility

The final simulator source and binary SHA-256 values are `0d7000b628257311515e8edd43492a348c9bdb3903c95c06bff5c718bf1f8a2` and `58c22764eb864f156bc1df4a0190a0ac1851ff2505fda5fe07b9fde2e4469c82`. Fine holdout seeds are 2026091701 (driven) and 2026091702 (equilibrium). Final model, protocol, command, raw-data, and output hashes are included in the experiment directory. The machine-readable evaluator is `scripts/final_ness_density.py`.